Geometry Unit 10 Assessment Guide

General Information	The benchmarks MA.912.GR.6.2 and MA.912.GR.6.4 build on a student's prior knowledge of circles and proportional reasoning. For students who need further support, consider using content for these Grade 7 benchmarks: MA.7.AR.3.2, MA.7.AR.4.5, MA.7.GR.1.3, and MA.7.GR.1.4. Students may choose to solve the problems using a variety of methods. Consider following student work to consider whether minor errors demonstrate a student's knowledge of the relationships in a circle. If a student consistently rounds too early in finding a solution, rounds to the wrong place value, provides an answer in the wrong form, forgets to include units, or repeats another common error, then consider the totality of the error and make a single overall deduction rather than multiple deductions.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.912.GR.6.4
area: 361.911 m^2		Recommended Scoring (6 Total Points)	Consider giving students 4 points if they show that they understand the proportional relationship but mistakenly use the diameter instead of the radius to find the area of the circle.
Question	**1b**	**Benchmark(s)**	MA.912.GR.6.4
arc length: 147.288 m		**Recommended Scoring** **(6 Total Points)**	Consider giving students 4 points if they show that they understand the proportional relationship but mistakenly use the diameter instead of the radius to find the arc length.
Question	**2a**	**Benchmark(s)**	MA.912.GR.6.4
arc length: 16.2π cm		**Recommended Scoring** **(6 Total Points)**	Consider giving students 3 points if they show that they understand the proportional relationship but mistakenly use the wrong angle measure or the wrong radius.
Question	**2b**	**Benchmark(s)**	MA.912.GR.6.4
arc length: 21π cm		**Recommended Scoring** **(6 Total Points)**	Consider giving students 3 points if they show that they understand the proportional relationship but mistakenly use the wrong angle measure or the wrong radius.

Question	3	Benchmark(s)	MA.912.GR.6.2
$m\angle KXW =$ 74°		**Recommended Scoring** **(6 Total Points)**	Students may choose to solve this problem using different methods. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error. Consider breaking down the points by verifying each of the following: • Correctly determines $m\,\widehat{KNW} = 254°$ • Correctly demonstrates the formula "measure of angle formed = $\frac{1}{2}$(major arc-minor arc)" • Correct angle measure

Question	4	Benchmark(s)	MA.912.GR.6.2
$x =$ 7		**Recommended Scoring** **(6 Total Points)**	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider breaking down the points by verifying each of the following: • Correctly determines the $m\,\widehat{RT} = 6x + 69$ • Correctly demonstrates the formula "measure of angle formed = $\frac{1}{2}$(difference of intercepted arcs)" • Correctly determines the value of x.

Question	5	Benchmark(s)	MA.912.GR.6.4
distance: 11.0 ft		Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider breaking down the points by verifying each of the following: • Correctly determines the $m\,\widehat{BT}$ • Correctly determines the $m\,\widehat{WB}$ • Correctly determines the $m\,\widehat{DN}$ • Demonstrates the understanding of the proportional relationship by correctly determining the length of $m\,\widehat{WB}$ • Demonstrates the understanding of the proportional relationship by correctly determining the length of $m\,\widehat{DN}$ • Correctly determines the distance
Question	6a	Benchmark(s)	MA.912.GR.6.2
$m\angle LBH =$ 52.5°		Recommended Scoring (6 Total Points)	No partial credit is suggested.
Question	6b	Benchmark(s)	MA.912.GR.6.2
$m\angle DWL =$ 90°		Recommended Scoring (6 Total Points)	No partial credit is suggested.
Question	6c	Benchmark(s)	MA.912.GR.6.2
$m\,\widehat{BW} =$ 55°		Recommended Scoring (6 Total Points)	Students may choose to solve this problem using different methods. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.

Question	7	Benchmark(s)	MA.912.GR.6.4
area | 40.218 cm^2		**Recommended Scoring** **(6 Total Points)**	Consider a deduction of 2 points for students who show that they understand the proportional relationship but make a minor error that is carried through, thereby resulting in a different answer.
Question	**8a**	**Benchmark(s)**	MA.912.GR.6.2
$x =$ | 6		**Recommended Scoring** **(6 Total Points)**	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error. Consider breaking down the points by verifying each of the following: • Correctly demonstrates the formula "measure of angle formed = $\frac{1}{2}$(difference of intercepted arcs)" • Correctly determines the value of x.
Question	**8b**	**Benchmark(s)**	MA.912.GR.6.2
$m\angle XWN =$ | 39°		**Recommended Scoring** **(8 Total Points)**	Consider following student work to determine if a minor error is carried through thereby resulting in a different answer. Consider deducting 2 points for such a minor error. Consider breaking down the points by verifying each of the following: • Correctly determines the $m\,\widehat{XN}$ • Correctly determines the $m\,\widehat{MR}$ • Correctly demonstrates the formula "measure of angle formed = $\frac{1}{2}$(sum of intercepted arcs)" • Correctly determines the measure of the angle

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<tr><th>Question</th><th>9a</th><th>Benchmark(s)</th><td>MA.912.GR.6.2</td></tr>
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$y =$	30

</td><th>Recommended Scoring

(6 Total Points)</th><td>Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error.</td></tr>
<tr><th>Question</th><th>9b</th><th>Benchmark(s)</th><td>MA.912.GR.6.2</td></tr>
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$m\angle KXC =$	$52°$

</td><th>Recommended Scoring

(8 Total Points)</th><td>Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error. Consider breaking down the points by verifying each of the following:
• Correctly determines the $m\angle XMC$
• Correctly determines the $m\widehat{XC}$
• Correctly demonstrates the formula "measure of angle formed = $\frac{1}{2}$measure of the intercepted arc"
• Correctly determines the measure of the angle</td></tr>
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